Balaji Rajesh Nandure

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.04
Swami Vivekanand Institute of Polytechnic, Latur	Class XII (MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION)	2019	89.00%
JAYKRANTI HIGH SCHOOL, LATUR	Class X (MAHARASHTRA STATE BOARD OF SECONDARY AND HIGHER SECONDARY EDUCATION, PUNE)	2016	83.00%

WORK EXPERIENCE

Ajapro AI Technologies

- Developed a **context-aware LLM-PostgreSQL conversational agent** using the **OpenAI**, and function calling to retrieve, interpret, and respond to multi-table data queries, automating **75% of manual lookups**.
- Implemented **Stripe subscription billing** with Silver, Gold, and Platinum tiers, integrating checkout, subscription lifecycle management, and webhook-based payment/status synchronization with automated billing workflows.
- Implemented federated **OAuth 2.0/OIDC** social logins (**Google, GitHub, LinkedIn, Facebook**) for seamless SSO.

PROJECTS

TransactiWar: Secure Full-Stack Banking Website [Demo]

- Built a secure **PHP/MySQL banking application** with account management, deposits, withdrawals, and transaction funcs.
- Implemented **defenses against SQLi, XSS, CSRF, LFI, and RFI** using input validation, session security, and access controls.
- Conducted **penetration testing with Burp Suite/Wireshark** and deployed the application in an isolated **Docker** container.

Auction Hub : Real-Time Auction Platform [Demo]

- Built a real-time auction platform using **React, Node.js, Express, MongoDB, and Socket.io** with live bidding notifications.
- Engineered Socket.io room-based bidding with automated closures, bid tracking, winner determination, and synchronization.
- Implemented **JWT/bcrypt auth and Stripe** checkout with protected APIs, payment workflows, and winning-bid processing.

CollabDoc : Real-Time Collaborative Editor [Demo]

- Developed a real-time collaborative document editor with **multi-user editing** via Operational Transformation and Quill Deltas.
- Engineered JWT/RBAC security, debounced MongoDB persistence, Socket.io rooms, and Stripe subscriptions.

WasteSort AI: Real-Time Conveyor Waste Detection & Tracking

- **Trained YOLOv11x** on 19 waste classes using progressive 640px and 1280px training, achieving **0.5695 mAP50**

(Rank 2).
- Improved small-object detection using **TTA, high-resolution inference, and optimized confidence/NMS thresholds**.
- Built **1024px/1280px ensemble tracking** with NMS fusion and centroid tracking, achieving **94.7% counting accuracy**.

3D2Fool: Qwen3.6 35B VLM-Based Defense Against Physical Adversarial Attacks agianst MDE

- **Reproduced 3D2Fool** using CARLA, PyTorch3D, and Monodepth2, trained printable 128×128 vehicle camouflage
- **Integrated Ollama with Qwen3.6 35B VLM** achieving **90% detection accuracy** over 150 CARLA test scenes.

RESEARCH

HyperLoRA: Continual Learning & Data-Free Unlearning for Generative & Discriminative LLMs

- Designed **HyperLoRA to mitigate catastrophic forgetting** in continual learning setup for LLM (Llama-3.2, Qwen2.5), achieving **81.5% retained accuracy** across 17 classification tasks, with **26.9% improvement** over sequential fine-tuning.
- Improved HyperLoRA for Llama-3.2B LLMs to freeze LoRA A and generate only LoRA B from 64-d task embeddings, **reducing parameters by 71.2%** from 153.8M to 44.3M and storage by 84.8% across 12 adapters as compared to per task lora.
- Developed **data-free machine unlearning** (GDPR right-to-erasure) without accessing forgotten data, achieving an zero-shot baselines (**-0.33%** residual forgetting) and audited privacy via **0.488 MIA AUC** against membership inference attacks.

SCHOLASTIC ACHIEVEMENTS

- **Ranked 2nd** at the IITH Waste Management Competition by engineering a robust **YOLOv8x pipeline**
- **Ranked 4th out of 176 teams** in the Brand & Buzzword Hackathon (Kaggle) with a **92.34%** final score.
- **Ranked 9th out of 74 teams** in the IITH Deep Learning Hackathon final leaderboard.
- **Ranked in Top 1500 nationally out of 50k+ competitors** in Google The Big Code across 3 programming rounds.

SKILLS

Programming Languages: C, C++, Python, JavaScript

Web Dev: Node.js, Express.js, React.js, Angular, WebSockets, OAuth, JWT, OpenAI API, Ollama, REST APIs, HTML5, CSS

Databases: PostgreSQL, MongoDB, Redis, Supabase, ChromaDB

AI/Ml: PyTorch, Transformers, RAG, Langchain, Scikit-learn, Pandas, NumPy, Statistics, Hugging Face, Unsloth, Matplotlib, PEFT

Developer Tools: Git, Docker, Postman, VS Code, Antigravity, Debugging

RELEVANT COURSES

Machine Learning, Deep Learning, Advanced Data Structures and Algorithms, Advanced Computer Networks, Network Security, Fraud Analytics Using Predictive and Social Network Techniques, Linear Optimization, Visual Computing

HOBBIES & INTERESTS

- Reading Books - Cooking Non-Vegetarian Dishes - Listening to Hip-Hop & Rap Music